

Logistic Regression: Motivation.

X... predictor (say, numerical for simplicity)

Y... response; categorical w/ two classes

$$Y = \begin{cases} 1 & \text{if category \#1} \\ 0 & \text{if category \#2} \end{cases}$$

Idea #1: $X \mapsto Y = \underbrace{\beta_0 + \beta_1 X}_{\in \mathbb{R}} + \varepsilon$ X

Idea #2: $X \mapsto \boxed{p(X) = \mathbb{P}[Y=1 \mid X]}$

$$= \underbrace{\beta_0 + \beta_1 X}_{\in \mathbb{R}} + \varepsilon$$
X

Def'n. $\boxed{\text{odds} = \frac{p(X)}{1-p(X)}} \in \underline{(0, \infty)}$

$$X \mapsto \text{odds} = \underbrace{\beta_0 + \beta_1 X}_{\in \mathbb{R}} + \varepsilon$$

Def'n. $\boxed{\text{logodds} = \ln(\text{odds}) = \ln\left(\frac{p(X)}{1-p(X)}\right)} \in \mathbb{R}$

$$X \mapsto \text{logodds} = \beta_0 + \beta_1 X + \varepsilon$$

⋮

$$\ln\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta_1 x + \varepsilon$$

$$\frac{p(x)}{1-p(x)} = e^{\beta_0 + \beta_1 x + \varepsilon}$$

$$\begin{aligned} p(x) &= (1-p(x))e^{\beta_0 + \beta_1 x + \varepsilon} \\ &= e^{\beta_0 + \beta_1 x + \varepsilon} - p(x)e^{\beta_0 + \beta_1 x + \varepsilon} \end{aligned}$$

$$\hat{p}(x) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 x}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 x}} \quad \leftarrow$$

$$\textcircled{Q}: \mathbb{P}[Y=0 \mid x] = \frac{1}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 x}}$$